"2024 - 07 -20"

Four taxa and two areas

Now, we load the data included in the package: tree and distribution.

```
## Libraries

library(blepd)
```

```
## Loading required package: ape
## Loading required package: picante
## Loading required package: vegan
## Loading required package: permute
## Loading required package: lattice
## This is vegan 2.6-6.1
## Loading required package: nlme
```

```
library(ggtree)
```

```
## ggtree v3.10.1 For help: https://yulab-smu.top/treedata-book/
##
## If you use the ggtree package suite in published research, please cite
## the appropriate paper(s):
##
## Guangchuang Yu, David Smith, Huachen Zhu, Yi Guan, Tommy Tsan-Yuk Lam.
## ggtree: an R package for visualization and annotation of phylogenetic
## trees with their covariates and other associated data. Methods in
## Ecology and Evolution. 2017, 8(1):28-36. doi:10.1111/2041-210X.12628
##
## Guangchuang Yu, Tommy Tsan-Yuk Lam, Huachen Zhu, Yi Guan. Two methods
## for mapping and visualizing associated data on phylogeny using ggtree.
## Molecular Biology and Evolution. 2018, 35(12):3041-3043.
## doi:10.1093/molbev/msy194
##
## Guangchuang Yu. Using ggtree to visualize data on tree-like structures.
## Current Protocols in Bioinformatics. 2020, 69:e96. doi:10.1002/cpbi.96
```

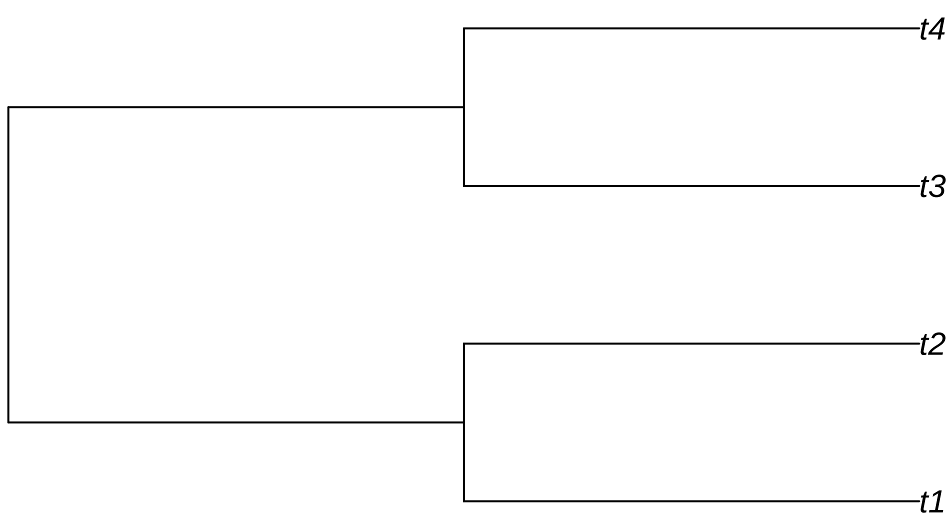
```
##
## Attaching package: 'ggtree'
## The following object is masked from 'package:nlme':
##
##     collapse
## The following object is masked from 'package:ape':
##
##     rotate
```

```
library(ggplot2)



## Data
```

```r
# Call the 4 terminals example

data(package = "blepd")

## trees

data(tree)

str(tree)
```

```
## List of 5
##  $ edge       : int [1:6, 1:2] 5 6 6 5 7 7 6 1 2 7 ...
##  $ edge.length: num [1:6] 1 1 1 1 1 1
##  $ Nnode      : int 3
##  $ tip.label  : chr [1:4] "t1" "t2" "t3" "t4"
##  $ root.edge  : num 1
##  - attr(*, "class")= chr "phylo"
##  - attr(*, "order")= chr "cladewise"
```

```r
plot(tree,use.edge.length =T)
```




```r
initialTree <- tree


## distributions
```

```r
data(distribution)

str(distribution)
```

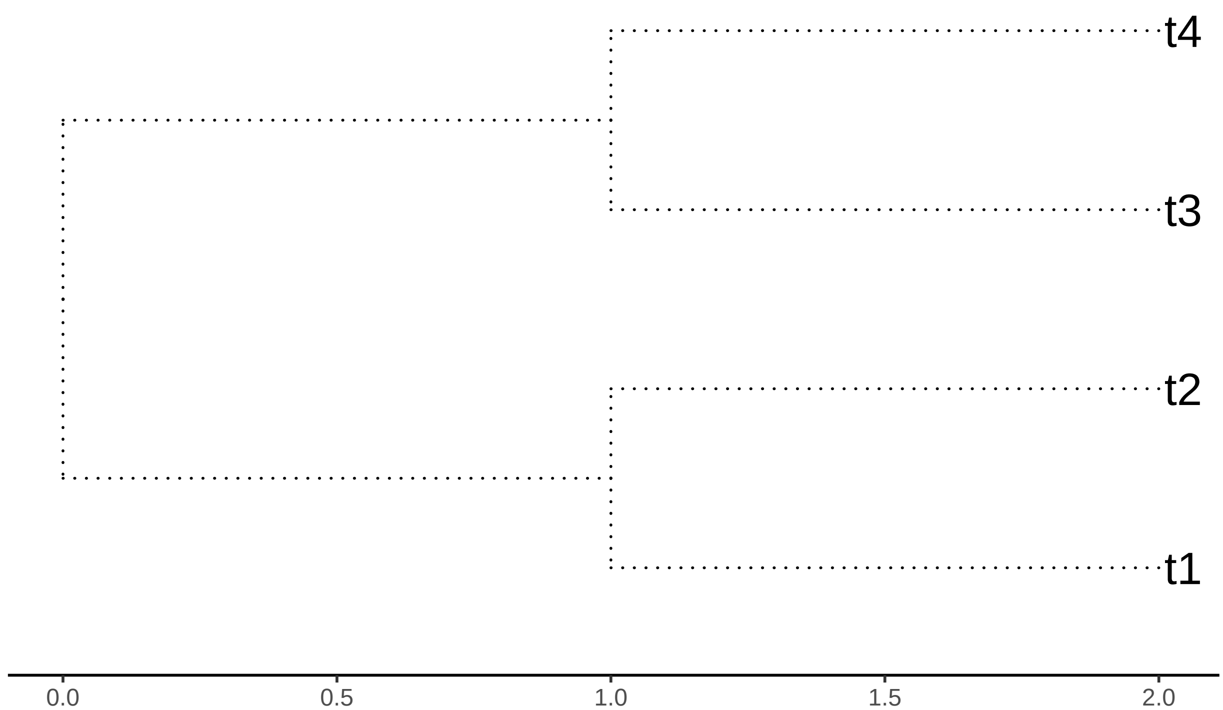
```
##  int [1:2, 1:4] 1 0 0 1 1 0 0 1
##  - attr(*, "dimnames")=List of 2
##   ..$ : chr [1:2] "A1" "A2"
##   ..$ : chr [1:4] "t1" "t2" "t3" "t4"
```

```r
dist4taxa <- distribution

## distribution to XY to be able to plot the data, otherwise, skip the next step

distXY <- matrix2XY(dist4taxa)

distXY
```

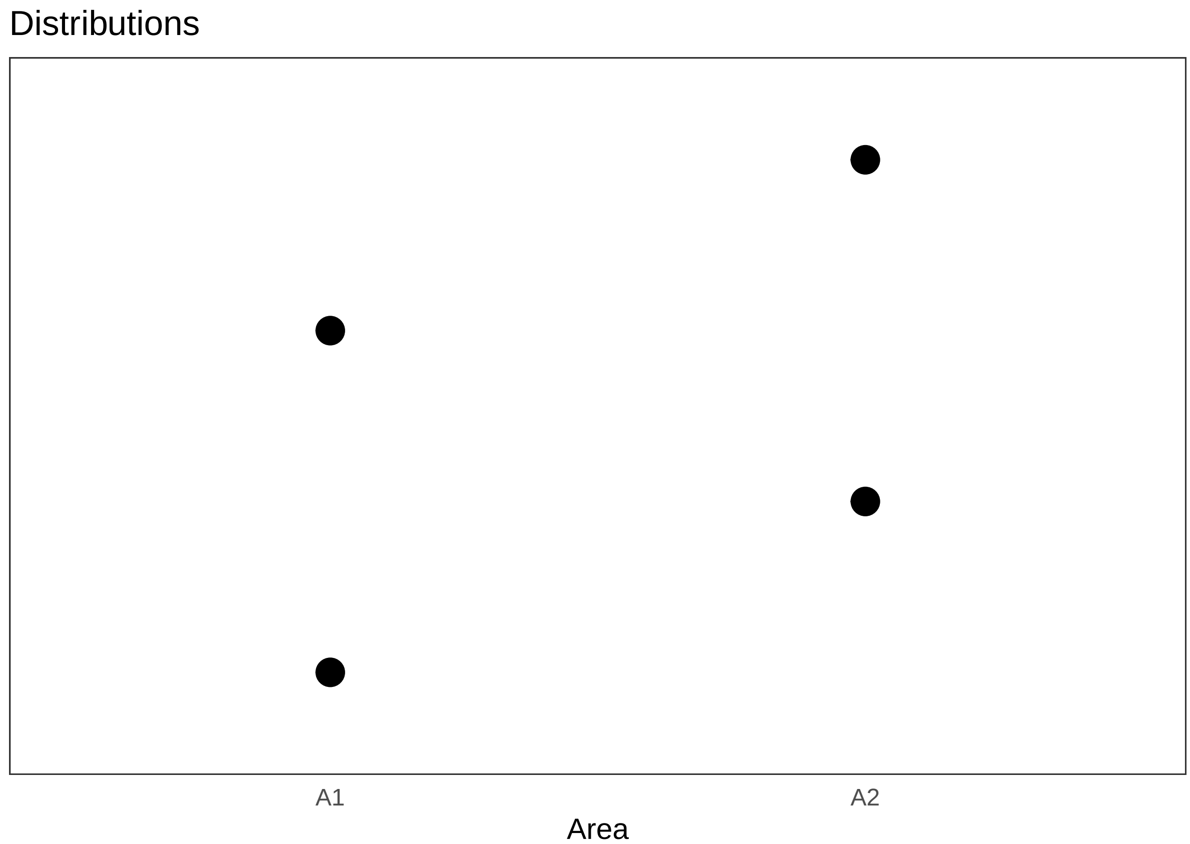
```
##   Terminal Area
## 1       t1   A1
## 2       t3   A1
## 3       t2   A2
## 4       t4   A2
```

```r
## plotting

## the tree

plotTree <-  ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted") +
             geom_tiplab(size=6, color="black") +
             theme_tree2() +
             ggtitle("Four terminals, equal branch length")


##
print(plotTree)
```

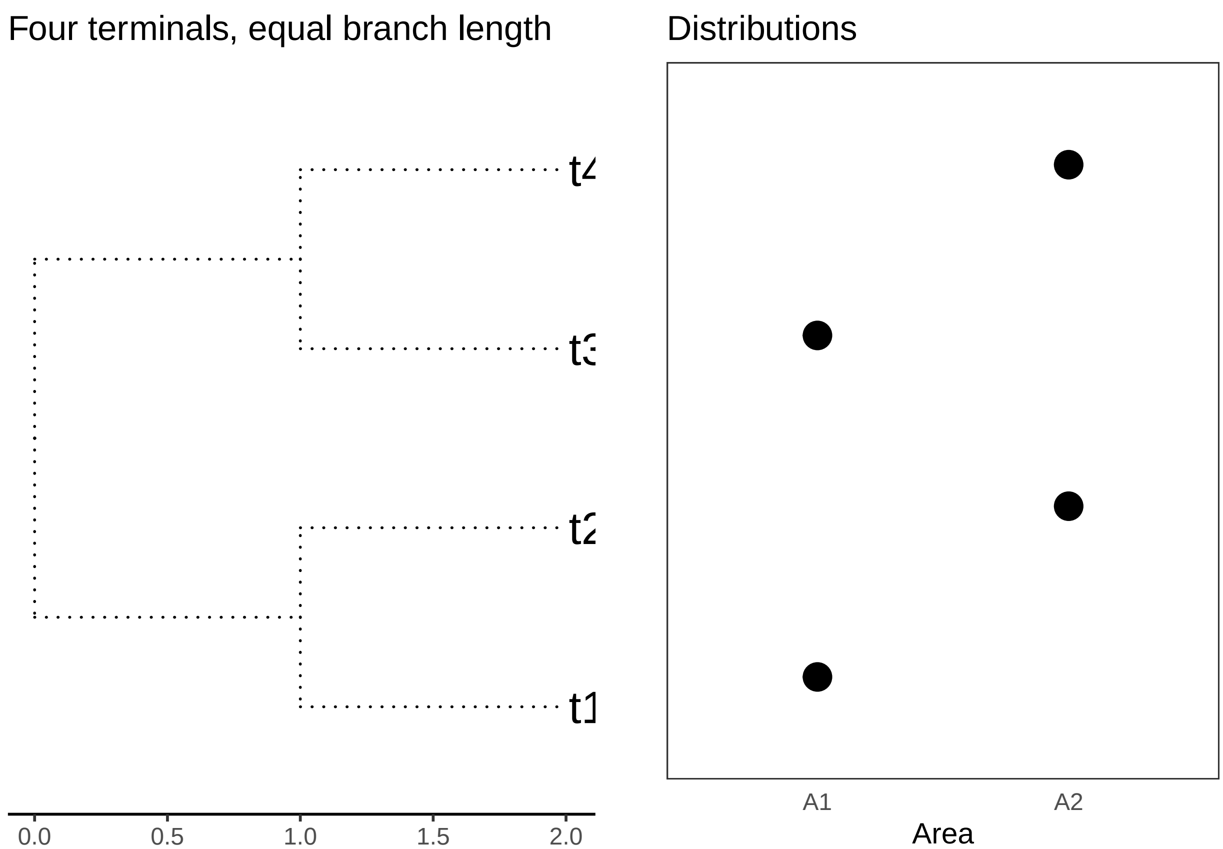
Four terminals, equal branch length




```r
## the distribution


plotDistrib <- ggplot(data=distXY,
                      aes(x= Area, y= Terminal , size =150))     +
               geom_point()                                      +
               theme_bw()                                        +
               theme( axis.text.y = element_blank(),
                      panel.grid.major = element_blank(),  # Remove major gridlines
                      panel.grid.minor = element_blank(),  # Remove minor gridlines
                      axis.line = element_line(colour = NA_character_),  # Remove axis lines
                      axis.ticks = element_blank(),  # Remove axis ticks
                      legend.position = "none"
                      ) +
               labs(title = "Distributions",
                    y = "",
                    x = "Area")


##
print(plotDistrib)
```




```
cowplot::plot_grid(plotTree, plotDistrib, ncol=2)
```



We check whether names in both objects: initialTree and dist4taxa, are the same.

```
##all(colnames(dist4taxa) == initialTree$tip.label)

all(colnames(dist4taxa) %in% initialTree$tip.label)
```

```
## [1] TRUE
```

We report the branch length, and calculate the PD values.

```
initialTree$edge.length
```

```
## [1] 1 1 1 1 1 1
```

```
initialPD <- PDindex(tree=initialTree, distribution = dist4taxa)

initialPD
```

```
## [1] 4 4
```

As expected there is a tie between both areas.

Now, we can see the impact of the posible null model(s), we will use the three available models to swap: “simpleswap”,“allswap”,“uniform”, and we will swap the three classes of branches: “terminals”,“internals”,“all”.

```
?swapBL

 for( modelo in c("simpleswap","allswap","uniform") ){

        for( rama in c("terminals","internals","all") ){
```

```r
            val <- swapBL(tree=initialTree,
                          distribution = dist4taxa,
                          model = modelo,
                         branch = rama
                         )

         cat( "\n\t Model=\t\t",modelo,
             "\n\t Branchs swapped=\t",rama,"\n" )

        printBlepd(val)

         cat( "\n\n\n" )

    }

  }
```

```
## model to test simpleswap reps 100
##
##   Model=      simpleswap
##   Branchs swapped=    terminals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1         A1A2  100     100
##
##
##
## model to test simpleswap reps 100
##
##   Model=      simpleswap
##   Branchs swapped=    internals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1         A1A2  100     100
##
##
##
## model to test simpleswap reps 100
##
##   Model=      simpleswap
##   Branchs swapped=    all
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1         A1A2  100     100
##
##
##
## model to test allswap reps 100
##
##   Model=      allswap
##   Branchs swapped=    terminals
```

```
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1         A1A2  100     100
##
##
##
## model to test allswap reps 100
##
##   Model=      allswap
##   Branchs swapped=    internals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1         A1A2  100     100
##
##
##
## model to test allswap reps 100
##
##   Model=      allswap
##   Branchs swapped=    all
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1         A1A2  100     100
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
##   Branchs swapped=    terminals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1         A1A2  100     100
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
##   Branchs swapped=    internals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1         A1A2  100     100
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
```

```
##   Branchs swapped=    all
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1         A1A2  100     100
```

As expected, as all braches are EQUAL nothing happens

Now, let us see if the results could be assigned to the longest branches (or not).

```r
## this is?
## source("/home/rafael/disco2/proyectosDRME/indices/blepd/docs/print00.r")


## we could use a single command (*evalBranch*)


?evalBranch

AllTerminals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval="terminals",
                           approach="all")
```

```
## Loading required package: maps

##
## Attaching package: 'phytools'

## The following object is masked from 'package:vegan':
##
##     scores
```

```r
str(AllTerminals)
```

```
## List of 2
##  $ lower:List of 4
##   ..$ :'data.frame': 1 obs. of  5 variables:
##   .. ..$ branchToEval: chr "[t1]|1"
##   .. ..$ resp        : chr "A1A2"
##   .. ..$ resp.1      : chr "A2"
##   .. ..$ approach    : chr "lower"
##   .. ..$ delta       : num -1
##   ..$ :'data.frame': 1 obs. of  5 variables:
##   .. ..$ branchToEval: chr "[t2]|2"
##   .. ..$ resp        : chr "A1A2"
##   .. ..$ resp.1      : chr "A1"
##   .. ..$ approach    : chr "lower"
##   .. ..$ delta       : num -1
##   ..$ :'data.frame': 1 obs. of  5 variables:
##   .. ..$ branchToEval: chr "[t3]|3"
##   .. ..$ resp        : chr "A1A2"
##   .. ..$ resp.1      : chr "A2"
##   .. ..$ approach    : chr "lower"
##   .. ..$ delta       : num -1
##   ..$ :'data.frame': 1 obs. of  5 variables:
##   .. ..$ branchToEval: chr "[t4]|4"
##   .. ..$ resp        : chr "A1A2"
```

```
##   .. ..$ resp.1      : chr "A1"
##   .. ..$ approach    : chr "lower"
##   .. ..$ delta       : num -1
##  $ upper:List of 4
##   ..$ :'data.frame': 1 obs. of  5 variables:
##   .. ..$ branchToEval: chr "[t1]|1"
##   .. ..$ resp        : chr "A1A2"
##   .. ..$ resp.1      : chr "A1"
##   .. ..$ approach    : chr "upper"
##   .. ..$ delta       : num 2.01
##   ..$ :'data.frame': 1 obs. of  5 variables:
##   .. ..$ branchToEval: chr "[t2]|2"
##   .. ..$ resp        : chr "A1A2"
##   .. ..$ resp.1      : chr "A2"
##   .. ..$ approach    : chr "upper"
##   .. ..$ delta       : num 2.01
##   ..$ :'data.frame': 1 obs. of  5 variables:
##   .. ..$ branchToEval: chr "[t3]|3"
##   .. ..$ resp        : chr "A1A2"
##   .. ..$ resp.1      : chr "A1"
##   .. ..$ approach    : chr "upper"
##   .. ..$ delta       : num 2.01
##   ..$ :'data.frame': 1 obs. of  5 variables:
##   .. ..$ branchToEval: chr "[t4]|4"
##   .. ..$ resp        : chr "A1A2"
##   .. ..$ resp.1      : chr "A2"
##   .. ..$ approach    : chr "upper"
##   .. ..$ delta       : num 2.01
```

```
#### so far so good

## this part below is strange

##############################################################



printEvalBranch(AllTerminals)
```

```
##     node initialArea FinalArea Aproach %Delta
## 1 [t1]|1        A1A2        A2   lower     -1
## 2 [t2]|2        A1A2        A1   lower     -1
## 3 [t3]|3        A1A2        A2   lower     -1
## 4 [t4]|4        A1A2        A1   lower     -1
## 5 [t1]|1        A1A2        A1   upper   2.01
## 6 [t2]|2        A1A2        A2   upper   2.01
## 7 [t3]|3        A1A2        A1   upper   2.01
## 8 [t4]|4        A1A2        A2   upper   2.01
```

```
AllTerminals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval="internals",
                           approach="all")

printEvalBranch(AllTerminals)
```

```
## [1] node         initialArea FinalArea    Aproach      %Delta
## <0 rows> (or 0-length row.names)
```

```
AllInternals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval="all",
                           approach="all")

printEvalBranch(AllInternals)
```

```
##       node initialArea FinalArea Aproach %Delta
## 2  [t1]|1        A1A2        A2   lower     -1
## 3  [t2]|2        A1A2        A1   lower     -1
## 5  [t3]|3        A1A2        A2   lower     -1
## 6  [t4]|4        A1A2        A1   lower     -1
## 8  [t1]|1        A1A2        A1   upper   2.01
## 9  [t2]|2        A1A2        A2   upper   2.01
## 11 [t3]|3        A1A2        A1   upper   2.01
## 12 [t4]|4        A1A2        A2   upper   2.01
```

This is the same